

different from today's mouse. Apart from this, the device itself was capable of hypertext linking, full-screen document editing, context-sensitive help, networked document collaboration, e-mail, instant messaging, even video-conferencing! Though Engelbart's organisation had to shut down in 1989 due to lack of funding, this one demo had stirred things up, especially for a company that saw its future as bleak in front of technology that could make paper obsolete.

PAPERWORK

Xerox, reasonably alarmed by the possibility of the demise of their paper based company in a future world where people could collaborate and work on digital documents from across continents, saw reason to invest into and control this new technology. This led to the setup of PARC (Palo Alto Research Centre) in 1970, hot on the heels of Engelbart's demo. Employees at PARC, some of the best computer scientists from across the country, had full freedom to work on their dream projects to build the future of computing – which is exactly what they did.

Alto and Smalltalk, Xerox's take on the first graphical computer and graphical user interface, were functional initially in 1973 and 1974 respectively, although they were improved upon for years to come. Alto featured full raster-based bitmap graphics at a resolution of 606 by 808, and other innovations like a dynamic mouse cursor, a word processor, a bitmap editor that are so familiar to us today. What was missing was a consistent user interface across these applications, which was brought about by Smalltalk. Almost everything you use today on a personal computer – tiled windows, scroll bars, icons – was first implemented here. The combination of Alto and Smalltalk was essentially the first personal computer. But it was a completely different company that realised its benefits.

THE FRUIT OF EDEN

By the time Xerox went commercial with their product, most of its good engineers and innovators had left for other brighter avenues. One such avenue was Apple, founded by Steve Jobs and Steve Wozniak in 1976. Although Apple's Lisa computer, which had started off as a command line based business-use computer, was one of the first beneficiaries of the influx of PARC people into Apple, it wasn't until the 1984 launch of the much more affordable and lightweight Macintosh that Apple truly reaped the benefits of the innovations from PARC.

The Macintosh shipped with a GUI – a welcome relief over command line text-based interfaces (Listen to the reactions of the audience at the famous 1984 demo: <http://dgit.in/1984Demo>).



The Xerox Alto - the display was oriented to represent an A4 size paper accurately

Apple was set to ride a wave of success that would thwart out any competition for a good decade to say the least. The remaining years of 1980s saw Microsoft collaborate with Apple to develop applications for the Macintosh, in turn launching their own platform Windows – which, along with its successor, eventually failed to catch the attention in the market. Nobody was in a hurry to leave the comfort of the world of Macintosh yet. Until the 90s arrived.

THE REDMOND WAVE

The 1990s saw many smaller but significant GUIs dwindle out to leave only Apple and Microsoft to duke it out in the market. Windows 3.0 arrived in May 1990, and with it came the applications that Windows was already building for Macintosh. Word and Excel were particularly popular, along with the tons of other 3rd party applications that could now run on Windows. By the time Windows 3.1.1 had hit the market, with GUI features like sculpted buttons, better color support, real-time multitasking, scalable TrueType font support and more, Windows-PCs were outselling Macs. The update, Windows NT, did

HOW IT ALL BEGAN



The first Macintosh

Back in the day, Apple provided a California elementary school free machines for all students. And during the summers, engineers from Apple worked with the teachers and kids to enhance the software and the GUI, mainly because they believed that their reactions would be the most honest.

similarly well and won over the business world too with good graphical support.

By this time, a rough lawsuit war had heated up between Apple and Microsoft over the design of Windows being similar to the Macintosh platform. But Microsoft turned the table with allegations that both Apple and Microsoft had merely taken the liberty of being inspired by technology from Xerox PARC. Microsoft won the lawsuit and the story hasn't changed much since then.

Apple eventually did up its game – the next gen OS X finally appeared in March 2001, along with the iMac line which breathed back some critical life into Apple's business.

GUIS RIGHT NOW

The GUI world hasn't changed drastically since the beginning of this millennium for the personal computer. On the other hand, the massively popular smartphone and its touch interface has brought about an entire new type of GUI. The next frontier in interfaces, virtual reality, demands something more from GUI. In a model of computing that can only be deemed successful if it replicates the real world accurately, the GUI has to resemble our real world as closely as possible – downright copying it if necessary. If someone in the world of tech does get it right, let's hope that this time, there won't be any lawsuits as a result. **d**